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Shallow Minds, Deep Concepts: AI's Struggle with Generics

I. Introduction

Generic statements, like “Tigers are striped” or “Mosquitoes carry the West Nile virus” are central to the way that we think, allowing us to make broad generalizations from limited data. These statements reflect some of our most deeply held assumptions about the world, showing causal structures and essentialized categories.

While generics are tools for understanding, they can also perpetuate harmful stereotypes. Philosopher Sarah-Jane Leslie differentiates these as *striking property generics*, such as “Pit bulls maul children,” or an even more extreme usage, “Muslims are terrorists.” Striking property generics spread negative stereotypes by attributing alarming or dangerous traits to entire groups based on minimal evidence, and can even be downright false. Despite their limited statistical prevalence, such generics are accepted as intuitively true, often influencing social judgments in ways that reinforce prejudice and discrimination (*Original Sin of Cognition*, p. 396).

The rise of artificial intelligence (AI), systems that rely on statistical classification, prompts the question: can AI replicate this human capacity to form and interpret generics? Upon first glance, AI uses classification systems that, like generics, uses data to make generalizations.

Moreover, AI systems such as generative adversarial networks (GANs) can produce outputs that mimic human-like generalizations.

The two corresponding images show an image and text based response of classic generics examples provided by ChatGPT. However, AI's reliance on statistical patterns rather than causal understanding raises doubts about their ability to authentically represent generics. As Leslie asserts, generics articulate cognitively fundamental generalizations shaped by

psychological factors (*Original Sin of Cognition*, p. 395). I argue that AI (perhaps luckily, in the extreme cases) cannot represent generics because it lacks the deep conceptual structures that underlie human cognition, depending instead on superficial statistical associations that mimic (but do not replicate) human reasoning. However, this theoretical limitation is also an ethical concern, as the deployment of AI systems based on shallow representations risks reinforcing the very biases that generic statements may often propagate (*Uncanny Performance, Divergent Competence*, p. 7).

II. Initial Understandings

Generics serve as a critical lens into human cognitive mechanisms. According to Leslie, they serve as language's way of articulating the "default generalizations," rooted in human cognitive tendencies to essentialize categories. From infancy, humans generalize features of a single item to others perceived as belonging to the same kind, often guided by language and

Please provide an image of a tiger

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Here is the image of a tiger you requested. Let me know if you'd like any adjustments or additional details!



Do mosquitos carry the West Nile virus (yes or no)



Yes.



innate assumptions about hidden properties (*Original Sin of Cognition*, p. 394). These generalizations, often guided by language, assume hidden properties that define kinds. Generics follow the structure *Fs are Gs*, where *F* denotes a kind term and *G* a feature term, such as *Tigers are striped* or *Mosquitoes carry the West Nile virus*.

Philosopher Hilary Putnam situates generics within the framework of natural kinds, categories of actual or potential of scientific importance (Putnam 1975, p. 140). Natural kind terms can demonstrate observable features as well as hidden essences that causally underpin those features, forming lawlike patterns essential to explanation and prediction (Putnam 1975, p. 229). For example, the natural kind term *tiger* assumes an essence that has stripes, roars, and is a predator.

In contrast, AI systems such as Artificial Neural Networks (ANNs) and GANs use statistical correlations rather than causal reasoning to group information. ANNs analyze vast datasets to extract patterns, while GANs generate outputs designed to mimic real-world phenomena. As Johnson and Dupre note, AI systems “squeeze all of the available information” out of training data but still lack the causal depth that characterizes human cognition (Johnson & Dupre, p. 2). A GAN system may produce realistic images of tigers by analyzing pixel-level patterns like stripes, but it cannot grasp the causal significance of a tiger’s predatory behavior or ecological role (Johnson & Dupre, p. 19).

III. On Generics

Leslie argues that generics articulate cognitively fundamental generalizations rooted in psychological essentialism, the human tendency to attribute hidden causal structures to larger categories. As Leslie explains, generic concepts have a unique truth value associated with them.

Even if they come off as intuitively true, it is possible that very few members of the kind actually possess the property. Their truth depends not on statistical prevalence but on causal and cognitive importance. She writes, “If even just a few members of a kind possess a property that is harmful or dangerous, then a generic that attributes that property to the kind is likely to be judged true” (Leslie 2017, p. 396). For example, while fewer than one percent of tigers eat people, this generic remains intuitively true because it references a property we have a strong interest in avoiding (*Original Sin of Cognition*, p. 396). Such generics enable us to generalize quickly and efficiently about potential threats or important features without requiring exhaustive statistical verification.

Leslie emphasizes that generics are shaped by our psychological essentialism, which assumes that certain categories, such as “tigers,” have underlying realities that cannot be directly observed. These essences are thought to causally explain observable features, like the stripes on a tiger, perhaps their rust-colored fur, or that they might find people to be a tasty dinner. While the features associated with an essence are neither necessary nor sufficient for category membership, they are nevertheless treated as predictive of the category’s essence (*Original Sin of Cognition*, p. 397). For instance, tigers are still considered members of the tiger-kind even if they lack stripes, yet stripes remain emblematic of the tiger essence. Perhaps more poignantly, tigers are still considered members of the tiger group even if they do not eat people.

IV. Main Argument

Johnson and Dupre lay out a central argument for why AI cannot represent generics is built on three premises:

- (1) There is a fundamental difference between “deep” and “superficial” (shallow) conceptual representations.
- (2) Human concepts are inherently deep, while AI representations remain shallow.
- (3) If human concepts are deep while AI representations are shallow, then a principled barrier to communication exists between human agents and AI.

First, there is a fundamental distinction between deep and shallow conceptual representations. Deep concepts go beyond initial observable features, using causal structures to explain and connect those features. This explanatory reasoning bridges the essence of a kind, or its underlying causal reality, to its observable properties. For example, the concept of *water* is a deep concept because it involves understanding that water’s molecular structure, H_2O , causally explains properties (liquidity, transparency, and its ability to quench thirst) (Putnam 1975, p. 241). Causal depth allows us to distinguish water from chemically or visually substances.

Putnam illustrates causal depth through the *Twin Earth* thought experiment (Putnam 1975, p. 230). Imagine a hypothetical planet, Twin Earth, identical to Earth in almost every way, including the presence of a substance that appears and behaves like water. Twin Earth’s “water,” however, is not H_2O but a different chemical compound, XYZ. The substances seem indistinguishable to the inhabitants of both planets. Putnam argues that the term *water* refers to H_2O on Earth and to XYZ on Twin Earth, not because of their superficial similarity but because reference is grounded in the causal structure of the substance’s essence. Thus, *water* on Earth and Twin Earth are distinct natural kinds, despite their identical appearances. This experiment demonstrates that deep concepts are rooted in the causal essence of a kind, rather than in its observable properties alone. Causal depth is thus essential to conceptual representation.

By contrast, shallow concepts are tied exclusively to surface-level information. AI systems rely on these shallow representations to classify data, identifying patterns among observable features without inferring causal relationships. As Johnson and Dupre argue, these representations depend on epistemological features rather than integrating causal or explanatory reasoning (Johnson & Dupre, p. 8). For instance, an AI tasked with classifying tigers may rely on superficial traits like stripes and fur patterns without grasping the ecological or behavioral significance of tigers, such as their role as predators or their relationship to humans.

Second, human concepts are inherently deep, while AI representations are shallow. Human cognition combines causal reasoning and psychological essentialism to infer unobservable properties under shared essences. Leslie reminds us that humans “attribute an underlying reality that explains the observable features of kinds” (Leslie 2017, p. 394). For example, humans generalize that *tigers eat people* not because most tigers exhibit this behavior, but because predation is essentialized as a causally relevant feature of the tiger kind. Through essentialism, we grasp the causal importance of striking properties like predation, which influence our perception of the kind as a whole (Leslie 2017, p. 396).

This depth is further evident in humans’ ability to override surface-level cues when causal structures conflict with appearances. Again consider Putnam’s *Twin Earth* thought experiment. Even without knowledge of molecular composition, humans distinguish between water (H₂O) and Twin Earth’s XYZ, understanding that their chemical essences differ. As Putnam notes, the term water refers to H₂O, not to anything that simply looks or acts like water (Putnam 1975, p. 241). This reliance on causal structures, rather than superficial traits, defines the depth of human concepts.

Conversely, AI representations are shallow, constrained by their dependence on statistical inference. ANNs and GANs can excel at identifying patterns among observable features, but their classifications are limited to surface-level correlations. Johnson and Dupre explain that AI systems do not engage with “causal-explanatory structures” (Johnson & Dupre, p. 11).

Third, if human concepts are deep while AI representations are shallow, then a principled barrier to communication exists between human agents and AI. Communication requires a shared conceptual framework, recognizing both the speaker’s intended sense and reference and the listener’s understanding. Putnam describes this alignment as necessary to linguistic meaning such that sense requires understanding the causal-explanatory framework underlying a concept, while reference ties this framework to real-world phenomena (Putnam 1975, p. 246).

AI systems, however, lack access to the causal depth that gives human concepts their meaning. Johnson and Dupre emphasize that this mismatch creates “critical miscommunications,” as AI terms differ from human concepts (Johnson & Dupre, p. 33). For AI, terms are statistical abstractions, reducible to patterns in the data. Humans, by contrast, embed these terms in context.

This divergence undermines the possibility of meaningful communication between humans and AI. When humans interpret generics, they rely on a shared framework of causal reasoning and psychological essentialism. AI’s reliance on statistical correlations, by contrast, precludes it from aligning its representations with human generics. Therefore, the lack of shared conceptual depth creates a principled barrier to communication, rendering AI incapable of authentically representing generics or understanding the nuanced messages humans intend to convey.

V. Counterfactual Dependency Objection

One objection to the argument that AI relies exclusively on shallow concepts is that ML systems could have the ability to detect counterfactual dependency, conditions under which one event depends on another. Counterfactual logic posits that if event A causes event B, then, in the absence of A, B should not occur. An ML system might thus infer causal relationships by observing patterns where such dependencies hold. For instance, it could detect that the absence of smoke strongly correlates with the absence of fire, potentially inferring a causal link between the two. At first glance, this capacity might appear to fulfill the requirements of causal reasoning, suggesting that AI could approximate the structures underlying generics.

However, counterfactual dependency does not necessitate causal reasoning. Leslie provides a compelling example to demonstrate the limitations of counterfactual logic. Imagine two kids, Billy and Sally throwing rocks at a bottle. Billy's rock strikes the bottle first, breaking it. However, had Billy not thrown his rock, Sally's rock would have broken the bottle instead. While the breaking of the bottle is counterfactually dependent on Billy's throw, it is not solely caused by his action. The causal explanation must consider the broader context, or "causal nexus," of the interaction (Leslie 2017, p. 398).

Generics depend on a cognitive framework that integrates causation with essentialism. Counterfactual dependency, while useful for identifying correlations, does not provide the causal depth required to navigate these frameworks. Leslie's example of striking property generics illustrates this point: while only a small fraction of mosquitoes carry the West Nile virus, the generic *Mosquitoes carry the West Nile virus* is judged true because of its causal significance and the salience of the property (Leslie 2017, p. 396). For humans, the causal link between

mosquitoes and disease transmission is essential to understanding the generic. AI systems, however, reduce such relationships to statistical associations, failing to account for their deeper causal structure.

Thus, while counterfactual analysis may superficially resemble causal reasoning, it remains insufficient for representing generics. AI systems cannot engage with the causal-explanatory depth that is integral to human cognition. As a result, AI is fundamentally limited to shallow representations, rendering it incapable of authentically representing generics or engaging in meaningful communication about them.

VI. Conclusion

Generics show the way we perceive the world, not as a collection of isolated facts, but as a web of interconnected causal structures. We instinctively search for meaning beneath the surface, guided by psychological essentialism and an innate drive to understand the “why” of things. AI, by contrast, views the world in fragments, piecing together statistical patterns without grasping the deeper threads that tie them together. This disconnect reveals more than just a technological limitation; there is a profound difference in the ways in which humans and machines understand and interact with the world (Johnson & Dupre, p. 18).

Generics involving social kinds, for instance, carry the weight of cultural assumptions and histories. When reduced to statistical abstractions, these generics risk perpetuating harmful stereotypes embedded in training data. Statements like *X group is violent* are no longer merely inaccurate; they become tools that reinforce biases, stripped of the nuance that human reasoning can provide. AI systems, incapable of distinguishing correlation from causation, lack the moral sensitivity to navigate such complexities.

AI's inability to represent generics is a reminder of what makes human cognition unique. We see beyond the visible; we infer, we question, we seek to understand the hidden connections that give the world coherence. While AI may excel in computation and prediction, it remains a mimic, unable to replicate the depth of human reasoning. Bridging this gap (although if we truly should remains open for debate) demands a rethinking of how we approach AI design, one that prioritizes causal reasoning and ethical safeguards. Until then, AI has yet to gain the ability to grasp the meaning beneath the surface.

Works Cited

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